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Model predictive control with dead-time compensation applied to a gas compression system

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ABSTRACT

The natural gas produced in primary separation is passed through a compression system to be pressurized and conditioned before being sent to its final destination. The operation of that system needs to be safe and efficient to avoid equipment damage and reduce energy consumption. The stable and secure operation of the equipment in a compression system is provided, many times, by a classical regulatory control layer. In this work, we present a Model Predictive Control (MPC) strategy to provide setpoints for the regulatory control layer of a gas compression system, aiming to avoid excessive energy consumption, decrease the plant variability, and guarantee a stable and safe operation against load disturbances. The proposed method is tested in a digital twin of a typical industrial unit using a Dead-Time Compensator Generalized Predictive Controller (DTC-GPC). Some disturbances in the feed flow rate of gas were considered as case studies. The controller responded satisfactorily to these disturbances keeping the plant operation stable and returning the controlled variables in the desired operating range after a short time.

1. Introduction

Natural gas obtained from primary separation in the offshore oil production platforms has to be conditioned and compressed before its destination, which can be natural gas processing units, gas-lift, fuel for turbines, or re-injection wells (Campos et al., 2013). The conditioning and compressing processes are made in a gas compression system where the gas is dried, sweetened, and compressed to a required delivery level (Mokhatab et al., 2018). The gas compression system can be single-stage or multistage with additional parallel trains sharing the load. The operational cost associated with those stages of compression based mainly on high energy-consuming centrifugal compressors is significant (Jones et al., 2017; Mohajer and Abbasi, 2017).

The correct operation and stability in the compression system are usually granted by controllers acting independently in the different equipment (Campos et al., 2013). In particular, the centrifugal compressor presents a limited normal operation range and is sensitive to operation changes. An instability condition causing that limitation in operation is denominated surge, when the compressor operates at high-pressure and a low mass flow rate. This phenomenon causes oscillation in the mass flow rate and pressure, undesired for a centrifugal compressor's safe operation. When the surge occurs in an uncontrolled way, it can cause massive reverse flow in the equipment, therefore, permanent mechanical damage. High flow rates may also take the compressor to a stonewall behavior, characterized by gas sonic velocity flows (Jones et al., 2017; Cortinovis et al., 2014; Torrisi et al., 2015; Daniarta et al., 2016). The surge condition may be avoided with the classical Proportional-Integral-Derivative (PID) controller or more advanced model predictive controller (MPC). For example, Cortinovis et al. (2014) implemented an anti-surge MPC that outperformed the conventional PID control strategy. Jung et al. (2017) presented an improved advanced feed-forward control structure to deal with both the poor load sharing that arises when the anti-surge controller becomes active in the slave compressor and when the anti-surge control has a time delay in low load operation.

Another important variable that is controlled locally in the

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compression system is the flared gas pressure. In this case, some classic controllers are commonly used in vessels such as headers to prevent pressure exceeding upper bounds against disturbances in the gas feed flow rate (Teixeira et al., 2010; Plucenio et al., 2017). The pressures in the different sections of the compression system are also controlled with regulatory controllers in the headers by manipulating the rotation speed of the compressors (Teixeira et al., 2010).

Classic and advanced control techniques have also been used in other upstream oil and gas industry processes, especially in offshore applications, to guarantee stability, safety, and increment the performance in operation (Krishnamoorthy et al., 2019; Strand and Sagli, 2004). MPC has gained much interest in those applications due to its advantages compared to classic controllers like the PID, it allows handling multivariable control problems naturally; it can take into account of actuator limitations; it allows operation closer to constraints, which frequently leads to more profitable operation (Maciejowski, 2002). An MPC application can be found in Ribeiro et al. (2016) where an MPC strategy was used to control a simulated complete production system that includes gas lifted wells, separation facilities, gas compression, and gas lift injection systems. Another example is the work of Diehl et al. (2019), where an advanced control strategy based on an MPC-PID coupling to handle instabilities in oil production wells was proposed. The MPC was used to deal with a slugging phenomenon while optimizing oil production. The controller was applied and validated in an ultra-deepwater field. This strategy was able to reduce the well down-hole pressure to increase oil production by approximately 10%. An MPC for sucker rod pumps was implemented in Hansen et al. (2019) for oil production systems. To estimate uncertain reservoir parameters, the authors used a Moving Horizon Estimation by only using common surface measurements.

Early detection of gas kick is another essential part of offshore platforms process safety. A kick is an undesirable event during drilling, workover, and completion operations caused by the influx of formation fluid into the wellbore when the pore pressure exceeds the bottom-hole pressure (Sule et al., 2019b; Bhandari et al., 2015; Muojeke et al., 2020). To mitigate these gas kick effects, advanced control techniques such as NMPC have been proposed in many works (Nandan and Imtiaz, 2016, 2017; Sule et al., 2019a; Amin et al., 2019). In these works, the NMPC responded satisfactorily, showing more robustness than PI or PID controllers to detect and attenuate swiftly kick disturbances. Additionally, in the work of Sule et al. (2019b), the necessity of modelling the influx fluid as a liquid-gas mixture to detect the kick event appropriately was shown.

Higher-level controllers acting more globally in a compressor system have hardly been reported in the literature. However, more general objectives can be pursued, such as minimizing nominal energy consumption while meeting the requirements of discharge pressures without affecting the plant's stability and reliability. For instance, when the compressor has high rotation speed, besides incrementing the operating cost, the machine tends to trip. That is an undesired condition because restarting the compressor may take about an hour. On the other hand, when the compressor is operated at a rotation speed below its nominal value, it will have high energy consumption to get the compressor back to the operational point when the gas inlet flow rate increases (Campos et al., 2013). Ensure the inter-stage differential pressure is also essential to avoid excessive variations in the discharge pressure and reduce the outbound pipelines vibrations. Abnormal vibrations can threaten the safety of pipelines (Liu et al., 2021). The surge condition and the flow of gas flared could be also controlled in a global way with MPC-PID structure (Plucenio et al., 2017). In that work, the controller structure was used to prevent surge, to reduce the flow of gas flared and to minimize an economic objective. However, in terms of practical implementation, it is unusual to modify the industrial compressor's surge control since it would lead to the loss of the equipment warranty by the machine's supplier.

This paper aims to present an application of control at a higher level for a gas compression system of offshore oil and gas platforms, seeking to minimize the modifications made to the plant and guarantee its stability. For the control purpose, a regulatory layer based on three PID controllers and a higher-level control layer based on an MPC responsible for manipulating the three setpoints of the regulatory system were implemented. The PID controllers are used to control the headers' pressure between sections. The MPC is tuned to control the power of each compressor and the pressure difference in each stage to maintain a narrow operating range that guarantees stability and reliability in operation. The gas compression system considered is based on a real multistage compression unit that contains different compression trains. This control application corresponds to a non-square system with more outputs than inputs. The role of MPC is to bring the plant to the desired operating zone, respecting the constraints of the process (Qin and Badgwell, 2003). For the gas compression system to operate in this manner, a soft-constrained operation must keep the controlled variables in the specifications (Shead et al., 2007).

The paper is structured as follows: in Section 2, the gas compression system used in this work is presented. In Section 3, the MPC algorithm implemented in the process is described. In Section 4, the MPC results in the gas compression system are presented and discussed, and finally, the conclusions are stated.

2. Gas compression system

The gas compression system considered in this work is a digital twin based on a typical industrial unit validated with some real data obtained from the Petrobras oil company. That unit is divided into three compression sections and modules between the first and second sections responsible for compressing and conditioning the gas (drying and sweetening) before final destination. The compression sections, which are defined as main, exportation, and re-injection are composed of trains of centrifugal compressors, turbines and other elements such as headers, scrubbers and heat exchangers. For the conditioning, membranes and molecular sieves are used.

The main compression section has two parallel trains of compression with a compressor on each train. The exportation section also has two trains of compression, with the difference that each train has two compressors in series. The re-injection section has only one train with one centrifugal compressor. The compressors are driven by turbines, where one turbine for each train is used, resulting in five turbines. At the beginning of each section, a header is arranged to regulate the pressure in each section. Additionally, at the inlet and outlet of each compressor, scrubbers that treat anti-surge recycling and heat exchangers that remove or supply energy are also used. The drying and sweetening modules are located just after the merging of the main section trains' outlet streams. For the sweetening that consists of removing gases such as CO_2 and H_2S, hollow fibre membrane contactors are considered. For the drying, where the moisture is removed, molecular sieves are considered. Besides, at the entry of both dehydration and sweetening modules, heat exchangers are also used. A simplified block diagram of the compression system is shown in Fig. 1. H_x blocks, with $x = 1, 2, 3$

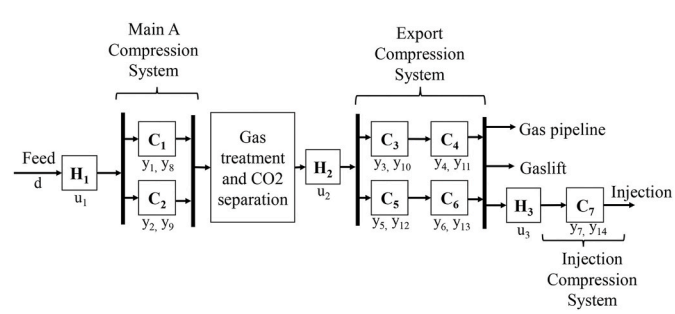

Fig. 1. Gas compressor plant block diagram.

represent the headers while C_y blocks with $y = 1, \ldots, 7$ represent the compressors.

Different regulatory controllers are implemented in the gas compression system to maintain the plant stability and control the pressure in the sections. Anti-surge PIDs controllers that keep the system stability are used to control the gas flow recycled in the compressors. The gas flow from the headers to the flare is controlled with PIDs that act on the relief valves. The surge condition and the flow of gas flared are controlled locally and are not considered as controlled/manipulated variables by the MPC as in Plucenio et al. (2017). In that work, the authors proposed an advanced control strategy acting on the regulatory control of a gas compression plant that manipulates the anti-surge control of the system, the flare relief valves and tries to minimize the energy consumption of the compressors. However, the industrial compressors, employed in this kind of process, have their internal anti-surge control included within the equipment warranty. A regulatory layer of PIDs controls the gas compression system pressure in the headers acting on the rotation of the compressor. As some variations of these last PIDs' setpoints are allowed without compromising the plant stability, they can be used as manipulated variables by the high-level MPC.

The mathematical models used to represent most of the equipment were based on mass and energy balances available in the open code library of EMSO (Environment for Modeling, Simulation, and Optimization) software, an equation-oriented process simulator (Soares and Secchi, 2003). In the case of the centrifugal compressors, a hybrid model was used. This model was based on phenomenological equations and real operation curves provided by the manufacturer. Those curves describe the performance maps referring to the polytropic head and polytropic efficiency. In the model, the performance curves are inserted in the form of second-degree multivariate polynomials. Each polynomial is a function of the inlet flow rate of natural gas and the rotation speed imposed on the compressor. For hollow fibre membrane contactors and molecular sieves, simplified algebraic models are used based on equipment efficiencies.

The models of all the equipment, including the PID controllers, were implemented in the 0.10.10 version of EMSO. The system of coupled differential and algebraic (DAE) equations was solved using the DAE system solver DASSLC that is included in EMSO (Secchi, 2012), with a relative accuracy of 10^{-6} and absolute accuracy of 10^{-8}.

3. DTC-GPC algorithm

The dynamic process model presents dead times, or time delays, mainly caused by the time required to transport mass and energy in the different sections. The decrease of the system phase caused by those time delays turns the control problem highly challenging. It is necessary to use adequate control techniques that incorporate a prediction of the process output, such as the Model Predictive Control (MPC) or the Dead-Time Compensator (DTC), to deal with this difficulty (Normey-Rico and Camacho, 2007).

MPC is an advanced control structure that uses a discrete technique to find the optimal control action by solving the open-loop optimization problem at each time step. This technique is based on the explicit use of a model to anticipate the outputs of the process at a future time and calculate the future control actions that minimize the problem cost function. Nevertheless, it is desired that the internal model of the MPC be as simple as possible, without significant loss of character of the process (Camacho and Bordons, 2007). That is done to avoid a complex optimization problem and reduce the algorithm's computational cost, which is mandatory for real implementation (Manenti, 2011).

DTCs are structures that use the system's dynamic model to allow, under some circumstances, the calculation of the control action considering the delay-free system. The most popular DTC is the Smith predictor, where one of its robust extensions is the filtered Smith predictor (FSP). This extension includes a low-pass filter, $\mathbf{F}_r(z^{-1})$, on the prediction structure to deal with the system uncertainties, mostly on time delays.

Due to the dynamic behavior of the gas compressors plant in this work, it was selected the Generalized Predictive Control in its dead-time compensation extension (DTC-GPC). MPC controllers based on linear models without guaranteed stability are the most common in the refineries and petrochemical plants. That is because the processes usually have stable open-loop behavior and the use of long prediction horizons and adequate weighting factors allows for stable closed-loop systems (Rossiter, 2004). Also, when the objective is to keep the process around the stationary state, the linear models provide good results when the plant is operating in the neighborhood of the operating point (Camacho and Bordons, 2007). The DTC-GPC uses the FSP structure into the MPC framework. It was demonstrated that substituting the implicit predictor structure of the GPC by an FSP could, with appropriate tuning, improve the robust stability of the design (Normey-Rico and Camacho, 2007). The prediction of the process output is split into two terms: (i) the FSP is used to compute the predictions until the dead time, and (ii) the MPC is used for the prediction from one sample after the dead time until the end of the prediction horizon.

Consider the discrete multivariable description of a process given by:

$$\mathbf{y}(k) = \mathbf{G}(z^{-1})\mathbf{u}(k), \tag{1}$$

where the variables $\mathbf{y}(k)$ and $\mathbf{u}(k)$ are, respectively, the $n_y \times 1$ output, the $n_u \times 1$ input vectors at time instant k. $\mathbf{G}(z^{-1})$ is a MIMO discrete $n_y \times n_u$ transfer function matrix with sampling period T_s and with the $g_{ij}(z^{-1})$ SISO transfer function element given by:

$$g_{ij}(z^{-1}) = \frac{z^{-1}B^*_{ij}(z^{-1})}{A^*_{ij}(z^{-1})} z^{-d_{ij}}, i = 1 : n_y; j = 1 : n_u \tag{2}$$

where d_{ij} is the dead time of the transfer function between the j-th input and i-th output expressed as a number of sampling times, and $B^*_{ij}(z^{-1})$ and $A^*_{ij}(z^{-1})$ are polynomials in the backshift operator, as detailed in Normey-Rico and Camacho (2007). The subscripts variation, $i = 1 : n_y$ and $j = 1 : n_u$, will be omitted in the rest of this work for the sake of readability.

It is important to note that every i-th output is influenced by an effective dead time given by $d_i = \min_j[\mathrm{d}_{ij}]$. Then, the input-output transfer function matrix can be expressed by the following rational matrix:

$$\mathbf{G}(z^{-1}) = \mathbf{D}(z^{-1})\widehat{\mathbf{A}}^{-1}(z^{-1})\widehat{\mathbf{B}}(z^{-1})z^{-1}, \tag{3}$$

where $\mathbf{D}(z^{-1})$ is a polynomial diagonal matrix with elements z^{-d_i}; $\widehat{\mathbf{A}}(z^{-1})$ is a diagonal polynomial matrix with elements $A_{ii}(z^{-1})$ equal to the least common multiple of the denominators corresponding to every row of the matrix $\mathbf{G}(z^{-1})$; $\widehat{\mathbf{B}}(z^{-1})$ is a polynomial matrix with elements $B_{ij}(z^{-1})$ such that:

$$\widehat{g}_{ij}(z^{-1}) = \frac{z^{-1}B_{ij}(z^{-1})}{A_{ii}(z^{-1})}, \tag{4}$$

$\widehat{g}_{ij}(z^{-1})$ is the SISO transfer function element without the common delay d_i of the corresponding output. The polynomials $A_{ii}(z^{-1})$ and $B_{ij}(z^{-1})$ are given by:

$$A_{ii}(z^{-1}) = 1 + a^{ii}_1 z^{-1} + a^{ii}_2 z^{-2} + \ldots + a^{ii}_{n_{a_i}} z^{-n_{a_i}}; \tag{5}$$

$$B_{ij}(z^{-1}) = b^{ij}_1 + b^{ij}_2 z^{-1} + b^{ij}_3 z^{-2} + \ldots + b^{ij}_{n_{b_{ij}}} z^{-n_{b_{ij}}}. \tag{6}$$

The DTC-GPC considers the Controlled Auto-Regressive Integrated Moving Average (CARIMA) model of the process and the disturbances to forecast the future behavior of the system. The CARIMA model can be expressed by:

$$\widehat{\mathbf{A}}\left(z^{-1}\right)\mathbf{y}(k)=\mathbf{D}\left(z^{-1}\right)\widehat{\mathbf{B}}\left(z^{-1}\right)\mathbf{u}(k-1)+\frac{\mathbf{C}\left(z^{-1}\right)}{\Delta\left(z^{-1}\right)}\mathbf{v}(k), \tag{7}$$

where $\mathbf{v}(k)$ is the $n_y \times 1$ noise vector at time instant k. $\mathbf{C}(z^{-1})$ is an $n_y \times n_y$ monic polynomial matrix representing the colouring polynomials of the noise and is usually considered as a design parameter (Tran et al., 2015). $\Delta(z^{-1})$ is the difference operator, i.e., $\Delta(z^{-1}) = 1 - z^{-1}$.

In this work, $\mathbf{C}(z^{-1})$ is considered as identity matrix. Since $\widehat{\mathbf{A}}(z^{-1})$ is diagonal, it is possible to use a diophantine equation to obtain the optimal predictions of each i-th output of the system as shown below:

$$A_{ii}\left(z^{-1}\right)y_i(k)=z^{-d_i}\widehat{\mathbf{B}}_i\left(z^{-1}\right)\mathbf{u}(k-1)+\frac{1}{\Delta}\mathbf{v}(k), \tag{8}$$

where $\widehat{\mathbf{B}}_i(z^{-1}) = [B_{i1}, B_{i2}, \ldots, B_{in_u}]$. The polynomial identity,

$$1=E_{ij}\left(z^{-1}\right)\Delta A_{ii}\left(z^{-1}\right)+z^{-j}F_{ij}\left(z^{-1}\right), \tag{9}$$

allows obtaining the following optimal prediction of the system:

$$\widehat{y}_i(k+n|k)=F_{ij}\left(z^{-1}\right)\widehat{y}_i(k)+z^{-d_i}E_{ij}\left(z^{-1}\right)\Delta\widehat{\mathbf{B}}_i\left(z^{-1}\right)\mathbf{u}(k-1), \tag{10}$$

where $E_{ij}(z^{-1})$ is of order $j-1$ and $F_{ij}(z^{-1})$ of order n_{ai}.

The various MPC algorithms propose different cost functions to compute the control action. The implementation of the DTC-GPC is based on the resolution of the following quadratic program optimization problem (QP):

$$\min_{\mathbf{u},\varepsilon_k}\left\{\mathrm{J}=\sum_{i=1}^{ny}\sum_{n=1}^{p}\left\{R_{ii}\left[\widehat{y}_i(k+n|k)-w_i(k+n|k)\right]^2\right\}+\right.$$

$$\left.\sum_{j=1}^{nu}\sum_{n=1}^{m}\left\{W_{jj}\left[\Delta u_j(k+n-1|k)\right]^2\right\}+\rho_\varepsilon\varepsilon_k^2\right\} \tag{11}$$

subject to:

$$u_{min,j}\le u_j(k+n-1|k)\le u_{max,j},\ n=1:m \tag{12}$$

$$\Delta u_{min,j}\le u_j(k+n-1|k)-u_j(k+n-2|k)\le \Delta u_{max,j},\ n=1:m \tag{13}$$

$$y_{min,i}-\varepsilon_k\le \widehat{y}_i(k+n|k)\le y_{max,i}+\varepsilon_k,\ n=1:p, \tag{14}$$

where p is the prediction horizon, and m is the control horizon. The future reference trajectory for i-th plant output at n-th prediction horizon step is given by $w_i(k+n|k)$ and the predicted value of i-th plant output at n-th prediction horizon step is given by $\widehat{y}_i(k+n|k)$, where k is the current time interval. Soft constraints were implemented to obtain a feasible QP solution for a non-square system with more controlled variables than manipulated variables. Those constraints are based on a dimensionless and nonnegative slack variable, ε_k, which quantifies the worst-case constraint violation. $\mathbf{R}$, $\mathbf{W}$, and ρ_ε are positive semidefinite weighting matrices.

The DTC-GPC predicts the output of the plant up to $k+d_i$ using the FSP and from $k+d_i+1$ to $k+p$, the prediction is computed using the normal procedure of the GPC (Normey-Rico and Camacho, 2007).

A block diagram of the DTC-GPC is shown in Fig. 2, where $\mathbf{G}_o(z^{-1})$ is the transfer function matrix without the effective dead time. From this diagram, it is possible to infer the prediction of the optimal predictor:

$$\widehat{\mathbf{y}}_p(k)=\mathbf{G}_o\left(z^{-1}\right)\mathbf{u}(k)+\mathbf{F}_r\left(z^{-1}\right)\left[\mathbf{y}(k)-\mathbf{G}\left(z^{-1}\right)\mathbf{u}(k)\right]. \tag{15}$$

where the filter $\mathbf{F}_r(z^{-1})$ is a free tuning parameter.

It is possible to calculate the predictions, from $k+d_i+1$ to $k+p$, that are used in the cost function J of Eq. (3) as a function of a free-response $\mathbf{f_r}$ (which depends on past actions) and the forced response of the system (which depends on future actions). For this, if the prediction of Eq. (10) is applied recursively, the output prediction of GPC control can be reduced to the following matrix expression:

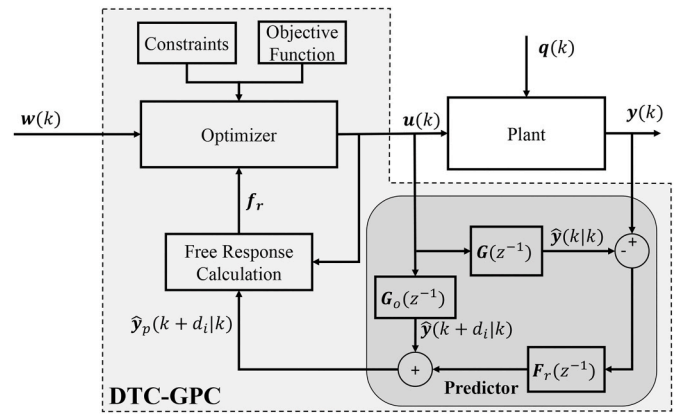

Fig. 2. General control structure of the DTC-GPC.

$$\widehat{\mathbf{y}}=\mathbf{H}\mathbf{u}_f+\mathbf{H}_p\mathbf{u}_p+\mathbf{S}\widehat{\mathbf{y}}_p, \tag{16}$$

where $\mathbf{H}\mathbf{u}_f$ is known as forced response, and $\mathbf{f_r}=\mathbf{H}_p\mathbf{u}_p+\mathbf{S}\widehat{\mathbf{y}}_p$ is a free response of the system, and:

$$\widehat{\mathbf{y}}=\left[\widehat{\mathbf{y}}_1,\widehat{\mathbf{y}}_2,\ldots,\widehat{\mathbf{y}}_{\mathbf{n_y}}\right] \tag{17}$$

$$\mathbf{u}_f=\left[\mathbf{u}_{\mathbf{f}_1},\mathbf{u}_{\mathbf{f}_2},\ldots,\mathbf{u}_{\mathbf{f}_{\mathbf{nu}}}\right]$$

$$\mathbf{u}_p=\left[\mathbf{u}_{\mathbf{p}_1},\mathbf{u}_{\mathbf{p}_2},\ldots,\mathbf{u}_{\mathbf{p}_{\mathbf{nu}}}\right]$$

$$\widehat{\mathbf{y}}_p=\left[\widehat{\mathbf{y}}_{\mathbf{p}_1},\widehat{\mathbf{y}}_{\mathbf{p}_2},\ldots,\widehat{\mathbf{y}}_{\mathbf{p}_{\mathbf{ny}}}\right]$$

$$\mathbf{H}=\begin{bmatrix}\mathbf{H}_{11} & \mathbf{H}_{12} & \cdots & \mathbf{H}_{1\mathbf{n_u}}\\ \mathbf{H}_{21} & \mathbf{H}_{22} & \cdots & \mathbf{H}_{2\mathbf{n_u}}\\ \vdots & \vdots & \ddots & \vdots\\ \mathbf{H}_{\mathbf{n_y}1} & \mathbf{H}_{\mathbf{n_y}2} & \cdots & \mathbf{H}_{\mathbf{n_y}\mathbf{n_u}}\end{bmatrix},$$

the future control actions are given by:

$$\mathbf{u}_{\mathbf{f_j}}=\left[\Delta u_j(k)\quad \Delta u_j(k+1)\quad \cdots\quad \Delta u_j(k+m)\right]^T, \tag{18}$$

and the past control actions are

$$\mathbf{u}_{\mathbf{p_j}}=\left[\Delta u_j(k-1)\quad \Delta u_j(k-2)\quad \cdots\quad \Delta u_j\left(k-n_{b_{ij}}\right)\right]^T. \tag{19}$$

Each block $\mathbf{H_{ij}}$ of the matrix $\mathbf{H}$ is calculated using the step response of the subsystem i,j, considering the polynomials $A_{ii}(z^{-1})$ and $B_{ij}(z^{-1})$.

The polynomial matrix $\mathbf{H_p}(z^{-1})$ has $(n_y p)$ rows and n_u columns and each element, $H^i_{p_{kj}}(z^{-1})$, is a polynomial with the same order as $B_{ij}(z^{-1})$. Here, i corresponds to the output, j to the input and k to the instant considered on the horizon, thus, $H^i_{p_{kj}}(z^{-1})$ gives the relation between $\Delta u_j(k)$ and $\widehat{y}_i(k+d_i+1|k)$.

Because of the independence between the predictions, $\mathbf{S}$ is a block-diagonal matrix, each block being of dimension $\sum_{i=1}^{n_y} p \times n_{a_i}$.

4. Results and discussion

The proposed controller maintain the compressors' powers (y_1, y_2, y_3, y_4, y_5, y_6, y_7) and the interstage differential pressures (y_8, y_9, y_{10}, y_{11}, y_{12}, y_{13}, y_{14}) around their nominal values to increase both the stability and reliability of the process. In the gas compression system, each header has a pressure PID controller acting on the rotation speed of the compressor to keep the pressure of each section at the desired value. The three PID pressure setpoints of these headers are the manipulated

variables of the MPC (u_1, u_2, u_3). The natural gas inlet flow rate is configured as a measured disturbance, which may eventually undergo positive and negative oscillations with relation to its the nominal value (q). The control system is composed of fourteen measured outputs and four inputs (14×4 MIMO system), three of them being the manipulated variables (control inputs) and the other one a disturbance. The controlled, manipulated, and disturbance variables can also be seen in Fig. 1.

The model that predicts and describes the dynamics of the compressor system for the MPC controller was implemented as a linear transfer function matrix near the operating point. The process variables are expressed in percentual deviations in relation to their nominal values, and were normalized to avoid scaling problems. The process data used to estimate the model parameters was obtained from disturbances applied in the digital twin running in the EMSO dynamic simulator.

The MIMO 14×4 transfer function matrix $\mathbf{G}(s)$, describing the process model, is given by Eq. (20). In that matrix, the first three columns correspond to the input-output models related to the manipulated variables and the last column to the disturbance models.

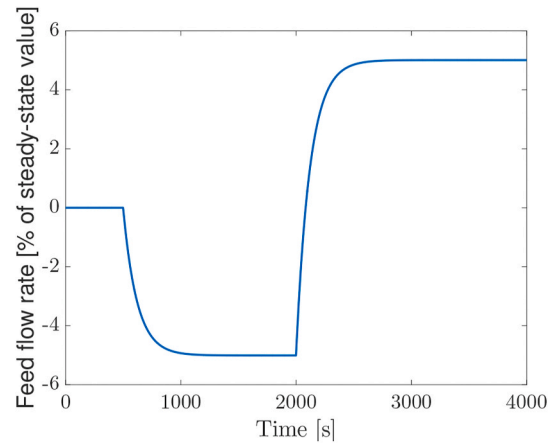

Fig. 3. Disturbances in the natural gas inlet flow rate, q.

$$\mathbf{G}(s) = \begin{bmatrix} \frac{-0.0391e^{-74s}}{s+0.0243} & \frac{0.0024e^{-78s}}{s+0.011} & 0 & \frac{40.65e^{-74s}}{s+16.6} \\ \frac{-0.0391e^{-74s}}{s+0.0243} & \frac{0.0024e^{-78s}}{s+0.011} & 0 & \frac{40.65e^{-74s}}{s+16.6} \\ \frac{(-0.018s-0.0005)e^{-74s}}{s^2+0.014s+0.0007} & \frac{-0.28e^{-74s}}{s+0.17} & \frac{0.006e^{-74s}}{s+0.012} & \frac{0.023e^{-74s}}{s+0.004} \\ \frac{(-0.028s-0.0005)e^{-74s}}{s^2+0.014s+0.0006} & \frac{0.438e^{-74s}}{s+0.251} & \frac{0.005e^{-74s}}{s+0.0110} & \frac{(0.053s)e^{-74s}}{s^2+0.01057s} \\ \frac{(-0.018s-0.0005)e^{-74s}}{s^2+0.014s+0.0007} & \frac{-0.28e^{-74s}}{s+0.17} & \frac{0.006e^{-74s}}{s+0.012} & \frac{0.023e^{-74s}}{s+0.004} \\ \frac{(-0.028s-0.0005)e^{-74s}}{s^2+0.014s+0.0006} & \frac{0.438e^{-74s}}{s+0.251} & \frac{0.005e^{-74s}}{s+0.0110} & \frac{(0.053s)e^{-74s}}{s^2+0.01057s} \\ 0 & 0 & \frac{-0.824e^{-74s}}{s+0.625} & \frac{(0.368s+0.0023)e^{-74s}}{s^2+0.02505s+0.001} \\ \frac{-0.0286e^{-74s}}{s+0.0269} & \frac{0.0112e^{-74s}}{s+0.015} & 0 & \frac{(45.34s+0.27)e^{-74s}}{s^2+19.09s+0.21} \\ \frac{-0.0286e^{-74s}}{s+0.0269} & \frac{0.0112e^{-74s}}{s+0.015} & 0 & \frac{(45.34s+0.27)e^{-74s}}{s^2+19.09s+0.21} \\ 0 & \frac{(-0.02s+0.0005)e^{-74s}}{s^2+0.039s+0.003} & \frac{0.0198e^{-74s}}{s+0.034} & \frac{-0.0004e^{-74s}}{s+0.001} \\ \frac{(-0.01s)e^{-76s}}{s^2+0.0366s} & \frac{-0.073e^{-74s}}{s+0.092} & \frac{0.023e^{-74s}}{s+0.025} & \frac{0.0049e^{-74s}}{s+0.0027} \\ 0 & \frac{(-0.02s+0.0005)e^{-74s}}{s^2+0.039s+0.003} & \frac{0.0197e^{-74s}}{s+0.034} & \frac{-0.0004e^{-74s}}{s+0.001} \\ \frac{(-0.01s)e^{-76s}}{s^2+0.0366s} & \frac{-0.073e^{-74s}}{s+0.092} & \frac{0.023e^{-74s}}{s+0.025} & \frac{0.0049e^{-74s}}{s+0.0027} \\ 0 & 0 & \frac{-0.0182e^{-74s}}{s+0.033} & 0 \end{bmatrix} \tag{20}$$

The performance of the proposed controller was evaluated against some positive and negative disturbances in the inlet feed flow rate of natural gas. The desired operating zones for the controlled variables were set to a range of ±5% of their steady state values while hard constraints restrict all the manipulated variables into a range of ± 10% of their steady state values. The maximum movement in the manipulated variables is ±0.5% of its perceptual scale. For the tuning of the controller the following parameters were used: the sampling time is $T_s = 2s$; the prediction horizon is $p = 100$; the control horizon is $m = 4$; the semidefinite weighting matrices are $\mathbf{R} = \mathbf{0}$ and $\mathbf{W} = \text{diag}[50, 1800, 4]$. The diagonal $\mathbf{R}$ matrix was set to zero since there are no fixed setpoints for the outputs, which may vary inside the soft constraints zones. The soft constraint violation penalty weight is $\rho_\varepsilon = 1000$, which gives the same importance to all controlled variables.

It was assumed in the simulation that all inputs and outputs were

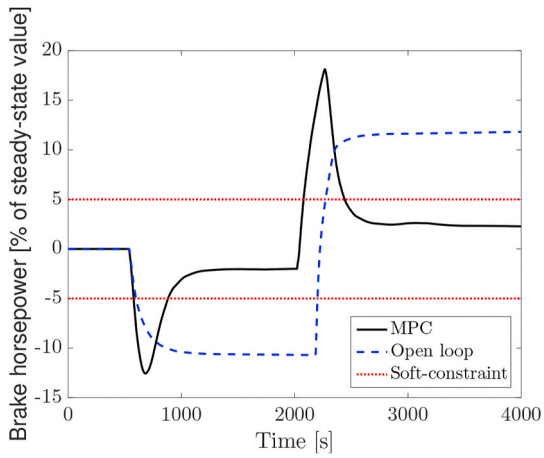

Fig. 4. Compressor consumption power C_1 in the main section with and without advanced control.

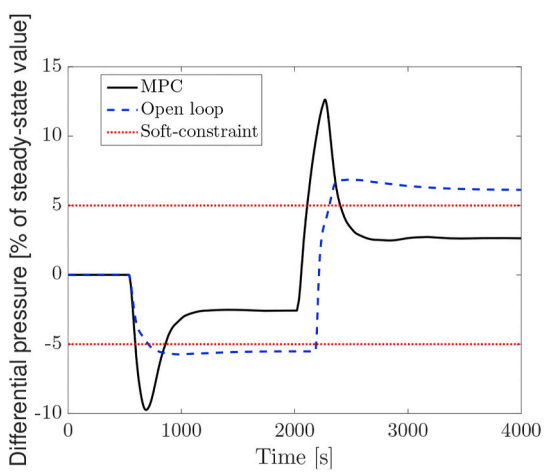

Fig. 5. ΔP_{C_1} differential pressure throughout compressor 1 in the main section.

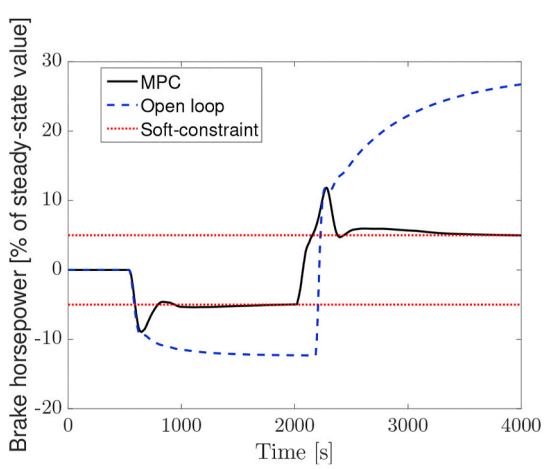

Fig. 6. Compressor consumption power C_3 in the exportation section with and without advanced control.

initially at their steady state values. At times t = 500s and t = 2000s, steps of −5% and 10% from the steady state, respectively, smoothed by a first-order dynamic, were imposed on the disturbance input of the system, q, as shown in Fig. 3.

The implementation of the model predictive controller is made using

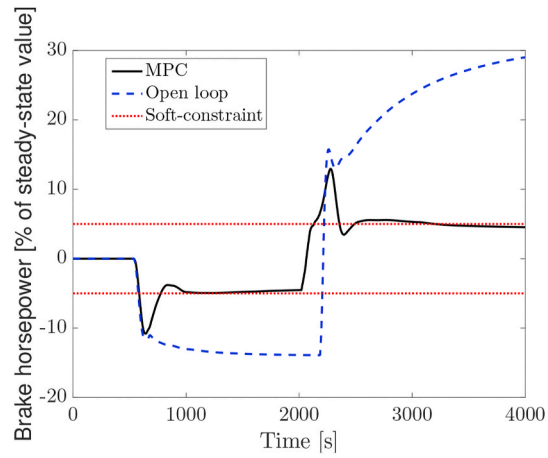

Fig. 7. Compressor consumption power C_4 in the exportation section with and without advanced control.

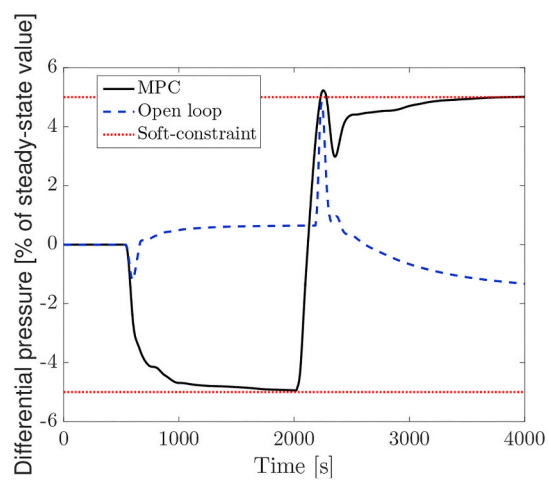

Fig. 8. y_{10}: ΔP_{C_3} differential pressure throughout compressor 3 in the exportation section.

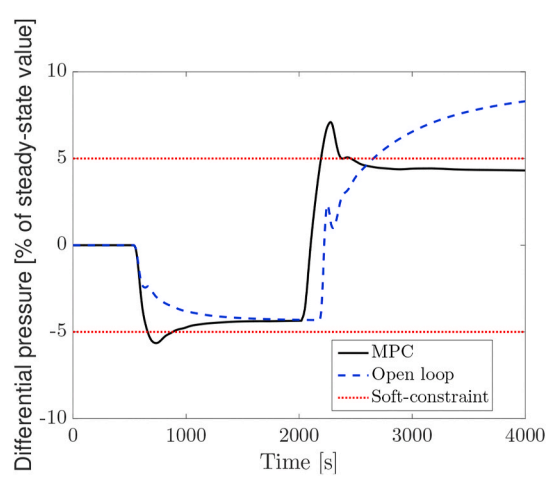

Fig. 9. y_{11}: ΔP_{C_4} differential pressure throughout compressor 4 in the exportation section.

MATLAB R2015b and Simulink. As the process simulations run in EMSO, a communication interface between Simulink and EMSO is used. In the digital twin, the dynamics in parallel trains are similar for each of the

sections with two trains, as shown in Fig. 1. That means that all variables in parallel behave the same way against the input disturbance. For instance, the power and the differential pressure in the compressors C_1 and C_2 are equal. Thus, the results are shown only for one of the compression trains in each section. The step responses of the output variables with the regulatory control without including MPC (open loop) and considering MPC are presented in Figs. 4–13.

It can be seen in Figs. 4 and 5 that the energy consumption and the differential pressure of the compressor C_1 in the main section change in the same direction as the disturbance q. After the first disturbance occurs, between t = 500s and t = 2000s, both the compressor power and the differential pressure of the compressor C_1 decrease due to a lower rotation speed of the compressor, imposed by the pressure regulatory control in the header H_1, resulting in less demand of energy. An increment in the percentage values in the output variables is observed after t = 2000s when q increases in the system. That increment is explained by the necessity to raise the compressor's rotation speed to process a higher load and reduce the header's pressure H_1. With the process in open-loop and the two disturbances, those variables leave their desired operation zones.

With MPC, the soft constraints are violated only for a short period after the disturbances are implemented. However, after that, the controller successfully manages to bring back the output variables into the desired zones by changing the regulatory layer setpoint of header H_1. That is a significant result because, with the possible operation margin of setpoint of header H_1, it is possible to keep the output variables in the desired range, despite the presence of disturbances.

In the Export section, keeping the output variables in the desired zone is more complex than in the Main section because the results are influenced by not only the control actions in the PID's setpoint of the header H_2 but also by the control actions in the setpoint of the header H_1. The results of compressor power and differential pressure for compressors C_3 and C_4 are presented in Figs. 6–9. The powers and the differential pressure percentage values are different for these two compressors in series for both the open-loop and with the MPC action. Analogously to the previous Main section, note that new setpoints for regulatory control are established when the MPC is activated. The advanced controller can bring the variables back within the desired operating zones.

The control of power and differential pressure in the Re-injection section is more challenging that in the other two sections, since the actions in the PID's setpoint of the headers of previous sections influence the result of this section. The behavior of the Re-injection compression system can be seen in Figs. 10 and 11. When the MPC is activated, only the soft constraints on the compressor power are violated for very short periods after the disturbance is implemented. This section shows the MPC's important contribution to keep the power of compressor C_7 (Fig. 10) in the desired zone. Also, an important energy reduction is noticed when the second disturbance was applied at t = 2000s. In the case of differential pressure of compressor C_7, the disturbances do not significantly affect the open-loop response as it is possible to see in Fig. 11 as the differential pressure C_7 remains in the desired operation region all the time.

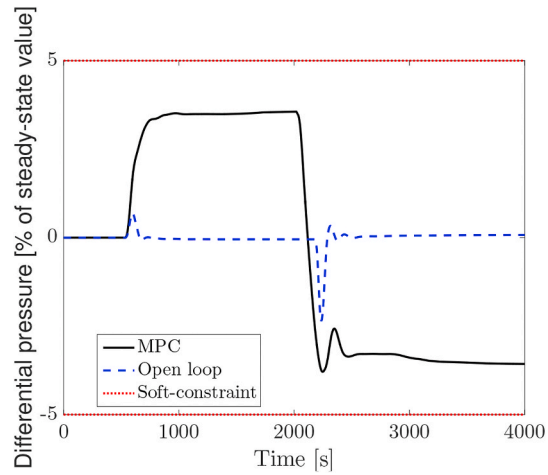

Fig. 11. y_{14}: ΔP_{C_7} differential pressure throughout compressor 7 in the re-injection section.

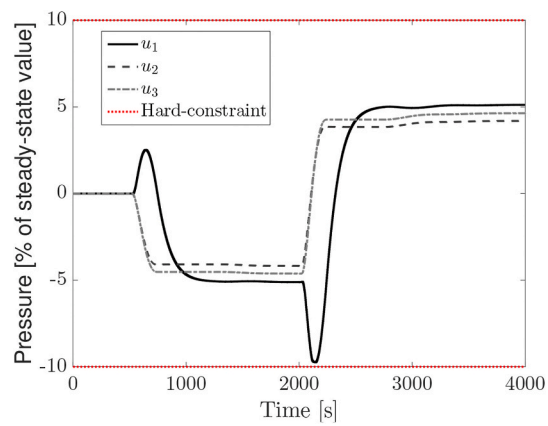

Fig. 12. PID pressure setpoints manipulated by the MPC controller.

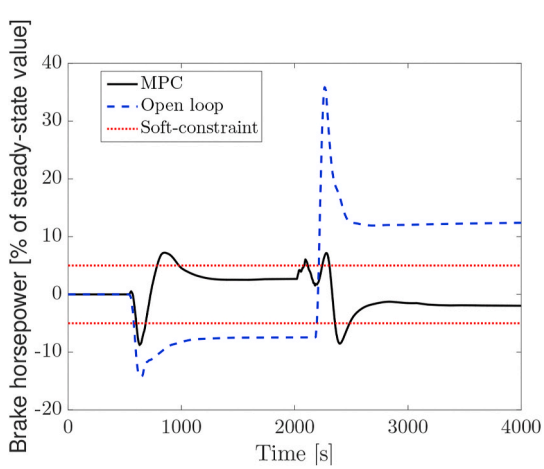

Fig. 10. Compressor consumption power C_7 in the re-injection section with and without advanced control.

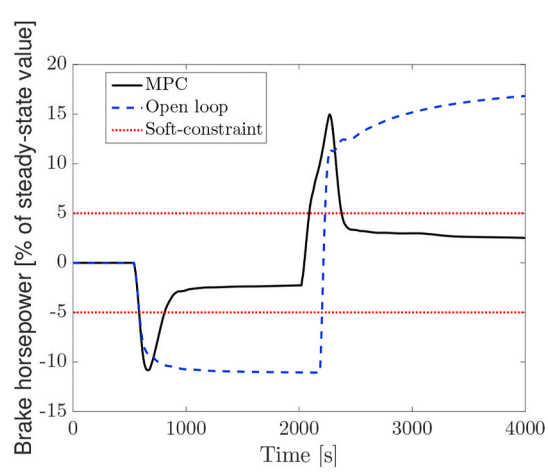

Fig. 13. Plant total energy consumption.

Fig. 12 shows the three control actions of the MPC applied to the regulatory PID controllers. The setpoint for the header H_1 regulatory control, given by u_1, follows an inverse response at the beginning of each disturbance and then follows the same direction. This direction change occurs when the power and the differential pressure of the compressor C_1 reach their minimal and maximal percentage values. The control action takes the system to a new steady state that is inside the desired control variable zones. In the case of the descendent disturbance, when the natural gas flow decreases, the header pressure decreases, and, consequently, the energy consumption in the Main section compression system decreases. Simultaneously, when the natural gas flow increases, the opposite effect is observed.

When the first disturbance is applied, the header H_2, governed by u_2, has its setpoint decreased by just under −4%. This behavior is justified since there is less gas coming from the Main A compression system. The compressor power increases slowly as already exposed by Figs. 6 and 7 to maintain the pressure balance at the output of the process. When the second disturbance is applied, the header H_2 pressure setpoint is increased to a new 4% steady state. Since more gas is entering the export stage, this pressure setpoint must keep the powers and differential pressures of both compressors close to the maximum limit and within the operating zones.

Similar behavior can be observed in the last header, H_3, which is manipulated by u_3, where the PID's setpoint decreases and increases in the same direction of disturbances to bring the controlled variables to the desired region. The most critical variable to control is the power since the pressure does not present significant changes in the open-loop response and remains inside the desired operating zone. The conjunct of the three control actions, to reject the disturbance, allows bringing the power of compressor C_7 back to a point close to the initial stationary state, as it is seen in Fig. 10.

In this implementation, the three control actions never reached the hard-constraints imposed on the PIDs controllers' setpoint values. That is a significant result because, with the appropriate headers' pressure setpoints, it is possible to guarantee that the controlled variables remain in their desired zones without compromising the plant safety.

The total energy consumption of the compressor system (all compressors) in both the open-loop and with the MPC action is presented in Fig. 13. In the open-loop case, significant energy consumption changes are seen in the same direction as the disturbances. When the incoming flow rate is low, between t = 500s and t = 2000s, the MPC manages to keep the energy consumption inside its desired zone around the steady state value. The energy consumption is higher than in the open loop case because the MPC was designed to prioritize stability over energy consumption reduction. After the second disturbance is implemented at t = 2000s, the controller was also able to keep total energy consumption in the desired region and the energy consumption had a percentage reduction of about 14% at the new steady state with regards to the open-loop. The total energy consumption of the compression system in all the time interval with the MPC action was reduced by 28% with regards to the regulatory control. The MPC control action showed that in addition to maintaining total energy consumption in the desired region and a more stable operation for negative oscillations in the natural gas flow rate, it should also help to reduce the risk of surge, as presented in Plucenio et al. (2017).

It is important to emphasize the potential of MPC application in gas compression systems. The case study considered here is based on a multi-variable problem with multiple constraints, where the complexity to keep the desired variables in the operating region increases in each section of the system since the control actions of the previous sections affect the subsequent sections. The MPC responded satisfactorily in this complex scenario and could bring economic benefits since it is possible to reduce the total energy consumption.

5. Conclusion

This work used an MPC as an advanced control layer over the gas compressor unit's regulatory control system. The gas processing plant was modeled and validated, considering phenomenological equations for most of the industry's equipment and real data. The MPC with the filtered Smith predictor extension was used in a complex scenario to maintain the power and the discharge pressure of compressors in desired regions.

The simplified model used by the MPC controller represented by a matrix of the transfer function of the process provided satisfactory predictions for the system's dynamics and time delays since the MPC managed to achieve its control objectives with good performance. It was possible not only to keep the compression plant operating in the desired region but also to reduce energy consumption despite positive disturbances of the feed flow rate of gas natural, compared to using only the regulatory control layer. The MPC showed the potential to be used as a supervisory controller in the compression plant. Important benefits can be obtained; such as the plant reliability and safety by guaranteeing that key variables remain into desired operating regions and to reduce operating cost by reducing the energy consumption.

Credit author statement

S.A.C. Giraldo: Methodology, Software, Validation, Writing – original draft, Writing – review & editing, Investigation. R.C. Supelano: Methodology, Validation, Writing – original draft, Writing – review & editing, Investigation. T.C. d'Avila: Methodology, Software, Validation. B.D.O. Capron: Methodology, Writing – original draft, Writing – review & editing. L.D. Ribeiro: Conceptualization, Resources, Supervision, Funding acquisition. M.M. Campos: Conceptualization, Resources, Supervision, Funding acquisition. A.R. Secchi: Conceptualization, Writing – original draft, Writing – review & editing, Supervision, Project administration

Declaration of competing interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

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